Homework 3 Quiz

- Due Mar 14 at 11:59pm
- Points 10
- Questions 11
- Available Mar 1 at 6:31pm - Mar 14 at 11:59pm
- Time Limit None
- Allowed Attempts 100

Instructions

Early Submission

Complete the quiz to get the 10pts for early submission.

Take the Quiz Again

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	13 minutes	6 out of 10

Correct answers are hidden.

Score for this attempt: 6 out of 10
Submitted Mar 14 at 1:02pm
This attempt took 13 minutes.

Question 1
1 / 1 pts
HW3P2: What is the loss that we use in this homework?

- MSE Loss
- Triplet Loss
- CTC Loss (selected)
- Cross Entropy

IncorrectQuestion 2
0 / 1 pts

HW3P2: How many mfccs are present in the train-clean-100 dataset?

28

Question 3

0 / 0 pts

HW3P2: Map the below given sequence of phonemes using CMUdict_ARPAabet.

['B', 'IH', 'K', 'SH', 'AA']

Do not put spaces between the labels.
Examples of valid answer formats:
abcde
[a,b,c,d,e]
['a,'b','c','d','e']

bIkSa

Question 4

1 / 1 pts

HW3P2: When using torchaudio's CUDA CTC Decoder, what should be the output of your model (i.e., the input to the decoder)?

Hint: Please refer to torchaudio's **CUDA CTC Decoder documentation (https://pytorch.org/audio/2.1/generated/torchaudio.models.decoder.cuda_ctc_decoder.html)** for more details

- ○ Wooden logs from Cedar trees
- ○ Raw logits
- ○ Probabilities (after applying softmax)
- ◉ Log probabilities (after applying log_softmax)

Question 5

1 / 1 pts

HW3P2: What is the shape of the **log_probs** required as input for CTCLoss? Consider input length as L, batch size as B, output classes as C.

- ○ C, B, L
- ○ L, C, B
- ◉ L, B, C
- ○ B, L, C

IncorrectQuestion 6

0 / 1 pts

HW3P2: In your starter code provided, you are asked to declare the decoder by using PyTorch Cuda CTC Decoder to decode phonemes

What does the `nbest` parameter in `cuda_ctc_decoder` control?

(**Hint:** check the docs: https://pytorch.org/audio/2.1/generated/torchaudio.models.decoder.cuda_ctc_decoder.html (https://pytorch.org/audio/2.1/generated/torchaudio.models.decoder.cuda_ctc_decoder.html))

- The number of best phoneme sequences to return for each audio input.
- The number of beams used during beam search decoding.
- The maximum length of the predicted phoneme sequence.

IncorrectQuestion 7

0 / 1 pts

HW3P1: In a Gated Recurrent Unit (GRU) cell, which of the following statements accurately describes the purpose of the reset gate?

- The reset gate helps the model decide whether to update the candidate hidden state based on the previous hidden state.
- The reset gate determines how much of the previous hidden state should be passed to the current time step.
- The reset gate controls how much of the current input should be combined with the previous hidden state.

Question 8

1 / 1 pts

HW3P1: In the context of natural language processing, what is the main purpose of using beam search during sequence generation?

- To find the highest probability sequence in a sequence generation model.
- None of the above.
- To optimize the model's parameters during training.
- To explore multiple possible sequences and maintain a set of top-K candidates.

Question 9

1 / 1 pts

HW3P1: Which of the following statements is incorrect?

- The derivative of an $N \times 1$ vector Y w.r.t an $M \times 1$ vector X is an $M \times N$ matrix.

- The derivative of a scalar L w.r.t an $N \times M$ matrix X is an $M \times N$ matrix.
- The derivative of a scalar L w.r.t an $N \times 1$ column vector x is a $1 \times N$ row vector ∇xL

Question 10

1 / 1 pts

HW3P1: What does CTC stand for in the context of neural network training for sequence-to-sequence tasks, and what is its primary purpose?

- CTC stands for "Cyclic Temporal Compression" and is employed to compress time-series data for efficient storage and processing.
- CTC stands for "Connectionist Temporal Classification" and is utilized for training sequence-to-sequence models where the alignment between input and output sequences is not one-to-one.
- CTC stands for "Comprehensive Textual Classification" and is designed for classifying complex textual data into multiple categories.
- CTC stands for "Characteristic Task Calibration" and is used for adjusting hyperparameters in deep learning models.

IncorrectQuestion 11

0 / 1 pts

HW3P1: For CTC decoding, what is not true about the use of blank symbol?

- It is not part of our SymbolSets.
- y-probs contains the probability that '-' occurs in each timestep as well as the probability of other symbols occurring at each time step.
- It can appear in our decoded result.
- It is represented as '-' in the writeup.

Quiz Score: 6 out of 10